Name: KIM DONGHWAN.

Matriculation No.: A0266094X.

Fumehood No: B7

Lab Group*: Monday *or* Thursday (*delete inappropriate)

Dates of Experiment:_____

CM3191 (AY2026/27 Semester 1)

INORGANIC Experiment 1

Homogenous Catalysis: Tris(triphenylphosphine)ruthenium(II) Dichloride – Synthesis, Characterization and Catalytic Studies

This lab proforma must be handwritten and its softcopy must be submitted together with the appendices (in ONE file) including the 1H-NMR, IR and ESI-MS spectra, of your products obtained, and your own datasheet(s) for all lab sessions, via Canvas-Assignments by 10:00 a.m. on 18-Sep-2026 (Fri.; week 6).

1. State the balanced reaction equation for the synthesis of $RuCl_2(PPh_3)_3$ in Session I.

$$2\,RuCl_3\cdot xH_2O + 7\,PPh_3 \xrightarrow{\text{dry EtOH, reflux}} 2\,RuCl_2(PPh_3)_3 + OPPh_3 + 2\,HCl + (2x-1)\,H_2O$$

Hallman, P.S.; Stephenson, T.A.; Wilkinson, G. Inorg. Synth. 1970, 12, 237-240.

2. Calculate the % yield of your $RuCl_2(PPh_3)_3$ obtained in Session I. *[Note: Detailed calculation steps are required.]*

Data: PPh_3 0.5076 g; $RuCl_3\cdot xH_2O$ 0.080 g; product = 1.895 − 1.653 = 0.242 g. M / g mol⁻¹: PPh_3 262.29; $RuCl_2(PPh_3)_3$ 958.85.
Equation: $2\,RuCl_3\cdot xH_2O + 7\,PPh_3 \rightarrow 2\,RuCl_2(PPh_3)_3 + OPPh_3 + 2HCl + (2x-1)\,H_2O$ ⇒ PPh_3 : Ru needed = 3.5 : 1; Ru : product = 1 : 1.

(a) Anhydrous basis — M($RuCl_3$) = 207.43 g mol⁻¹ (as instructed: lecture notes p. 57, Sigma-Aldrich 206229 "anhydrous basis")

n(PPh_3) = 0.5076 g ÷ 262.29 g·mol⁻¹ = 1.935×10^{-3} mol.

n($RuCl_3\cdot xH_2O$) = 0.080 g ÷ 207.43 g·mol⁻¹ = 3.857×10^{-4} mol.

Limiting reagent PPh_3 : Ru = $1.935\times10^{-3} / 3.857\times10^{-4}$ = 5.0 (>3.5) ⇒ $RuCl_3\cdot xH_2O$ limiting

Theoretical mass = 3.857×10^{-4} mol × 958.85 g/mol = 0.3698 g.

% yield = 0.242 g ÷ 0.3698 g × 100(%) = 65.4(%)

(b) Trihydrate basis — M($RuCl_3\cdot3H_2O$) = 261.47 g mol⁻¹ (x = 3, the usual literature formula, e.g. Inorg. Synth. 1970, 12, 237)

n(PPh_3) = 1.935×10^{-3} mol

n($RuCl_3\cdot3H_2O$) = 0.080 g ÷ 261.47 g/mol = 3.060×10^{-4} mol

Limiting reagent PPh_3 : Ru = $1.935\times10^{-3} / 3.060\times10^{-4}$ = 6.3 (> 3.5) ⇒ $RuCl_3\cdot3H_2O$ limiting.

Theoretical mass = 3.060×10^{-4} mol × 958.85 g mol⁻¹ = 0.2934 g

% yield = 0.242 g ÷ 0.2934 g × 100 = 82.5 %.

Conclusion / basis used:

65.4 ~ 82.5 %.

3. Interpret all the signals obtained in the ESI-MS spectra for your $RuCl_2(PPh_3)_3$ in Session I.

No molecular ion is seen
$RuCl_2(PPh_3)_3$ readily dissociates PPh_3 in solution and loses Cl^- under ESI condition
The base peak m/z 263 is $[PPh_3+H]^+$ and m/z 279 is $[OPPh_3+H]^+$
the phosphine oxide by-product of Ru(III)/Ru(II) reduction.
The Ruthenium-containing cluster at m/z 661, $[RuCl(PPh_3)_2]^+$ ($M-Cl+PPh_3$)
and m/z 625, $[Ru(PPh_3)_2-H]^+$ (i.e. 661 − HCl), show the unique wide envelope
expected from the seven Ru isotopes (96, 98, 99, 100, 101, 102, 104)
with additional +2 contribution from ^{37}Cl in 661-cluster
confirming the Ru-Cl-$(PPh_3)_2$ core.
The weak m/z 339 is tentatively $[PPh_4]^+$

4. Tabulate and assign the IR signals obtained for your $RuCl_2(PPh_3)_3$ in Session I.

No.	ṽ obs. / cm⁻¹	Int.	Assignment (vibrational mode)	Lit. / cm⁻¹ [ref.]
1	3048	weak	ν(C-H) aromatic	3058 (JCE)
2	2106	very weak	overtone?	
3	1570	weak	ν(C=C) ring.	~1585
4	1479	medium	ν(C=C) ring	1480 (JCE)
5	1431	strong	ν(C=C) ring + P-C(Ph)	1432 (JCE)
6	1186, 1155	weak.	in-plane δ(C-H)	1180, 1155
7	1085	med.	X-sensitive q band. ν(P-C(aryl))	~1090
8	1027	weak.	in-plane δ(C-H)	1027
9	999.	weak.	benzene.	998
10	741.	strong	out-of-plane γ(C-H), 5 adjacent H.	742 (JCE)
11	691.	v.strong	ring out-of-plane deformation	693 (JCE)
12	617	weak.	in-plane ring deformation	~617
13	571, 542, 517	med.	X-sensitive y/t modes	500~520
	<500	X	ν(RuCl) 300~320, out of measured range.	Nakamoto
	3368, 1588	X.	O-H, H_2O of $RuCl_2 \cdot xH_2O$ not observed.	

Int.: vw/w/m/s/vs. Ref.: [JCE] Miecznikowski et al., J. Chem. Educ. 2011, 88, 657 (same compound, ATR); Whiffen, J. Chem. Soc. 1956, 1350; Nakamoto, Infrared and Raman Spectra of Inorganic and Coordination Compounds, Part B.
All bands are those of coordinated PPh_3; O–H/H_2O bands of $RuCl_3 \cdot xH_2O$ absent; Ru–Cl and Ru–P stretches lie below 500 cm⁻¹.

5. Use no more than **100** words to comment the ^{1}H-NMR signals obtained in Session II and Session III.

[Notes: (i) Do NOT tabulate your experimental results. (ii) Only the first 100 words will be marked if more than 100-word answer is given.]

Both spectra were referenced to the mesitylene internal standard (CH$_3$ singlet set to 2.26 ppm; Ar-H 6.72 ppm)

Session II shows essentially unreacted 1-phenylethanol (CH quartet 4.77, CH$_3$ doublet 1.42, phenyl 8.2 – 7.4 ppm), only a trace acetophenone ortho-H doublet (7.90 / 7.88 ppm) and diacetone alcohol singlets (1.19, 2.06, 2.50 ppm) from K$_2$CO$_3$-promoted aldol-condensation of acetone.

Session III shows same signals, but acetophenone ortho-H doublet is roughly 5-times larger relative to the standard, and additional aromatic multiplets (7.3 – 7.6 ppm) from PPh$_3$/OPPh$_3$ release by the catalyst appear; the acetophenone COCH$_3$ singlet overlaps the diacetone alcohol CH$_2$ signal.

6. Calculate the % yields of acetophenone obtained in your Session II and Session III. Is there any implication found from such two results? Explain. *[Note: Detailed calculation steps are required.]*

Session I.

$n(I) = 0.07\ ml \times 0.801\ g/ml \div 120.19\ g/mol = 5.03 \times 10^{-4}\ mol$

n(1-phenylethanol charges) $= 0.06\ ml \times 1.012\ g/ml \div 122.17\ g/mol = 4.97 \times 10^{-4}\ mol$

$n(\text{acetophenone}) = \frac{0.0412}{2.7463/3} \times 5.03 \times 10^{-4} = 0.0224 \times 5.03 \times 10^{-4} = 1.13 \times 10^{-5}\ mol.$

% yield $= (1.13 \times 10^{-5} / 4.97 \times 10^{-4}) \times 100 = 2.3\%$

Session III (with catalyst)

$n(\text{acetophenone}) = \frac{0.1902}{2.6641/3} \times 5.03 \times 10^{-4} = 0.1086 \times 5.03 \times 10^{-4} = 5.41 \times 10^{-5}\ mol.$

% yield $= (5.41 \times 10^{-5} / 4.97 \times 10^{-4}) \times 100 = 10.9\%$

Implication

without $RuCl_2(PPh_3)_3$ only ~2% acetophenone is found. This is the clue to air oxidation (i.e. acetophenone impurity found in commercial 1-phenylethanol; air-oxidation only)

with catalyst, the yield is 5-fold. thus $RuCl_2(PPh_3)_3$ is a genuine (phe) catalyst for the hydrogen transfer oxidation.

(approx.) 16% loss of substrate carbon was same regardless of presence/absence of Ru-(phe) catalyst. This may be due to non-reaction related procedure, such as rotavap workup or transfer, etc...

7. Calculate the TOF of the $RuCl_2(PPh_3)_3$ in the catalytic reaction studied. *[Note: Detailed calculation steps are required.]*

$$TOF = \left\{ \frac{n(\text{product})}{n(\text{catalyst})} \right\} / \text{time (min)}$$

$n(RuCl_2(PPh_3)_3) = 0.060\,g / 958.85\,g\,mol^{-1} = 6.26 \times 10^{-5}\,mol$

$n(\text{acetophenone}) = 5.61 \times 10^{-5}\,mol$ (session II)

$t = 60$ min.

$\Rightarrow$ TOF $= 0.89$ /min.

* considering session II.

$$\left\{ \frac{(5.61 - 1.12) \times 10^{-5}}{6.26 \times 10^{-5}} \right\} / 60\text{ min} = 0.119\text{ /min.}$$

$\therefore$ 0.119 per minute.

($\approx$ 0.69 per hour)

8. Use no more than **80** words to comment the value of TOF obtained in Question 7, compared with literature value, and propose ONE possible for such TOF difference.

[Note: Only the first 80 words will be marked if more than 80-word answer is given.]

calculated TOF is below the 3 mol per hour TOF reported for the same reaction (~12 mol%) (J. Chem. Educ. 2011, 88, 687) One probable cause could be partial catalyst oxidation due to air and moisture. This is supported by observed greenish colour in reaction mixture as well as $OPPh_3$ reported by ESI-MS.

END OF LAB PROFORMA